

Recent advances and intensifications in Haber-Bosch ammonia synthesis process

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ABSTRACT

Ammonia is crucial as it serves as a key nitrogen source in fertilizer production to enhance crop growth and as an emerging energy carrier due to its high hydrogen content and ease of liquefaction. Despite various technological changes proposed and implemented since its inception, the Haber-Bosch process remains the predominant method for ammonia production. We first give a bird's eye view of current ammonia synthesis technologies available based on the latest trends, to justify why we think the conventional Haber-Bosch process is still a relevant technology worth investigation for further improvement. We review the engineering design modifications within the ammonia synthesis loop, examining improvements in the efficiency of ammonia synthesis. This review gives an overview of recent research and advancements focused on process intensification within the loop and its individual key components, i.e., the reactor and the catalyst, separation, and purge gas recovery technologies. It highlights significant progress and explores potential future directions in these areas.

1. Introduction

In the early twentieth century, Fritz Haber and Carl Bosch developed a process known as the Haber-Bosch process, which allowed for the large-scale production of ammonia [1]. The initial patent for ammonia synthesis was published by Fritz Haber and Robert Le Rossignol [2] in 1916. This development had a profound impact on society, initiating the first chemical global revolution. Ammonia has since been widely used in fertilizer manufacturing, contributing to the significant population growth over the past century [3]. Global ammonia production capacity estimated to stand around 188 million metric tonnes and is expected to grow worldwide by 1.54 % annually [4].

Chemical storage systems, particularly those involved in generating and storing hydrogen via water electrolysis, present a promising option for long-term energy storage. Such systems enable the effective integration of intermittent renewable power sources into the energy grid, providing a compelling alternative to short-term, limited-capacity solutions like batteries. Despite these advantages, the storage and transportation of hydrogen pose considerable challenges that need to be addressed. Currently, hydrogen is stored using cryogenic condensation to liquid form at 20–30 K, 5–10 bar, or compressed at 350–700 bar and room temperature [5]. Regrettably, these methods are hindered by low

gravimetric energy density, primarily due to the requirement of heavy storage systems, as well as challenges related to flammability and diffusivity of hydrogen. Consequently, these elements present practical limitations on the utilization of hydrogen as a medium for energy storage [6]. Ammonia is now being studied as a potential solution for long-term energy storage, going beyond its traditional uses as a fertilizer [7]. Ammonia has a larger energy density (12.7 MJ/L_{LHV} at -33°C) compared to liquid hydrogen (8.5 MJ/L_{LHV} at -253°C) [8], and its storage costs 26–30 times less than hydrogen [9]. The conversion of hydrogen to ammonia has emerged as a viable carbon-free hydrogen storage and transportation method [10].

Ammonia can be used directly as a fuel, which potentially results in a high production of NO_x [11]. Ammonia decomposition on the other hand offers a promising method to reduce NO_x emissions, with a mild endothermic reaction ($\Delta H_r = 46.22$ kJ/mol) [12]. Efforts have focused on developing efficient catalysts, with ruthenium being the most effective metal for ammonia decomposition. However, due to its high cost, alternative catalysts based on non-noble metals like iron, cobalt, and molybdenum are being explored [13].

Ammonia is a major commodity chemical with an existing global infrastructure for distribution [14,15] and regulation in place. If successfully deployed, its demand could increase exponentially, given the

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world's energy needs and the projected growth of renewable energy production [16]. Many nations such as Japan, Australia, Netherlands, and the United Kingdom are among those who have crafted clear plans towards using green ammonia as an energy storage material for renewable electricity surplus [17]. However, despite the promise of ammonia's second revolution, its current production method, mainly through the Haber-Bosch process using fossil fuels as feedstock, raises sustainability concerns. It is associated with drawbacks including elevated greenhouse gas emissions and considerable energy usage of 30 MJ/kg NH_3 [18]. These challenges stem from both reactant production and the energy-demanding, high-pressure, high-temperature synthesis of ammonia. This process is a significant contributor to greenhouse gas emissions – it produces 1.8 % of global anthropogenic CO_2 emission [19]. Ammonia production is reported to be twice as emissions-intensive as crude steel production and four times that of cement [20]. The CO_2 emission also depends on the technology and the feed being used to produce ammonia, ranging from 1.9 (t/t NH_3) for steam reforming of natural gas to 16.7 (t/t NH_3) for partial oxidation of coal [21].

The thermocatalytic processes of ammonia synthesis have reached a level of maturity (>100-year history) comparable to the petroleum refining and petrochemical industries. As mentioned earlier, ammonia synthesis holds potential on a large scale in the energy sector for energy storage and utilization, both as an energy carrier for hydrogen fuel and as a carbon-free fertilizer [22,23]. The energy transition by green ammonia must be supported by facilitating the downscaling of the process and robustness of the operations [24]. Indeed, smaller ammonia manufacturing operations can be attractive for its use as a locally consumed fertilizer or to fuel tractors and other equipment used in farming activities [23]. However, using the Haber-Bosch process at small scales increases energy consumption due to heat losses [3,25], but more significantly, increases the investment costs per unit of ammonia production. Large-scale operations, which are in the order of typically billions of euros, will make the adoption of this technology at a small scale a huge challenge [24].

Other than using inexpensive renewable energy resources [23], decarbonized ammonia manufacturing at smaller scales to reduce energy consumption can be achieved by implementing process intensification methods [22] in the existing Haber-Bosch process to increase its efficiency [23] and decrease the plant's cost. In the former approach, the hydrogen and nitrogen produced as the feed gas for the Haber-Bosch process are obtained using renewable energy. For instance, nitrogen can be separated from air in pressure swing adsorption (PSA) unit using electricity generated from wind, while hydrogen can be separated from

water via electrolysis using solar energy or produced by biomass gasification. Indeed, using renewable energy sources has the added benefit of reducing the carbon footprint [23]. Nonetheless, the cost of the overall ammonia production process using the Haber-Bosch process from hydrogen and nitrogen made using renewable energy to be twice as high as those produced from fossil fuels [26]. In this review article, we will not discuss engineering technologies of hydrogen and nitrogen production, but we will be reviewing the efforts in the latter approach on the progress made in the conventional thermocatalytic ammonia production using hydrogen and nitrogen in the Haber-Bosch process, with particular emphasis on the developments related to the key components of the process, aiming to highlight the efficiency improvements from these initiatives.

2. Decarbonizing ammonia: an overview of existing and advances in ammonia production technologies

A typical ammonia production plant, as depicted in Fig. 1, involves several key steps [27]. Although there are variations, particularly in the synthesis gas purification steps, the general process begins with the desulfurization of natural gas. This is followed by catalytic steam reforming, which is used to generate H_2 and CO . Then, through the water gas shift reaction, additional H_2 is produced and CO converts to CO_2 . The CO_2 is then removed through absorption or adsorption methods. Next, catalytic methane formation is employed to eliminate any remaining CO and CO_2 from the H_2 . N_2 is typically sourced from the air through air separation processes such as cryogenic distillation. The synthesis gas initially has insufficient pressure and needs additional compression before being introduced into the ammonia synthesis loop. The high-pressure synthesis gas is then fed into the ammonia reactor to produce ammonia through the Haber-Bosch process. Finally, in the separation step, the ammonia is separated from the unreacted gas.

Decarbonizing ammonia can be achieved through various approaches which can be categorized into three generations [28]. Generation 1 involves capturing and storing CO_2 emissions from conventional production, which also is known as “blue ammonia” in the colored scheme. Generation 2 uses renewable, rather than fossil fuel sourced, hydrogen to produce ammonia which is also called “green ammonia”, where carbon emissions can be eliminated from the synthesis gas production. Hydrogen production via steam methane reforming (SMR), for the Haber-Bosch process, is the main contributor ($\approx 90\%$) to the CO_2 emission of the whole process [29,30]. Decoupling ammonia production from SMR is challenging due to the higher costs and lower efficiency of

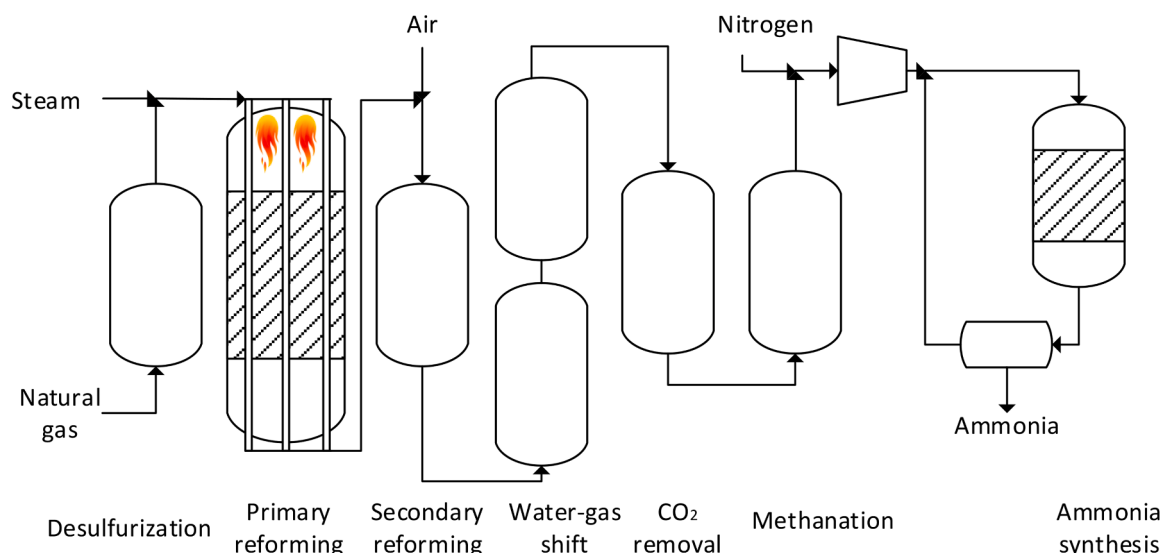


Fig. 1. Conventional ammonia plant scheme.

hydrogen production via water electrolysis compared to SMR. However, the declining cost of renewable electricity is making water electrolysis more competitive, and it is expected to become the dominant method for producing hydrogen [31]. The production of green hydrogen, particularly via water electrolysis powered by renewable energy sources, presents a substantial opportunity for reducing the carbon footprint associated with ammonia synthesis [32]. Additionally, water electrolysis yields higher-purity hydrogen, thereby eliminating the need to purge gas in the ammonia synthesis loop to reduce the accumulation of inert gases. Furthermore, high-pressure hydrogen production via electrolysis can enhance the energy efficiency in the ammonia synthesis loop. This process could minimize the energy requirements for compression to achieve the operating pressure necessary for the ammonia reactor. Also, an integrated process for ammonia synthesis using Solid Oxide Electrolysis (SOE) has been explored [33]. This approach achieves high energy efficiency by harnessing the steam generated by the synthesis reaction's heat and exploiting the favorable thermodynamics of high-temperature electrolysis. As outlined by Smith et al. [3], carbon-free ammonia production can be achieved using the traditional Haber-Bosch process, as long as it is decoupled from SMR, electric compressors replace steam turbine compressors, and alternative catalyst and ammonia separation techniques are employed to decrease operating pressure.

Biological nitrogen fixation occurs naturally in certain microorganisms, such as nitrogen-fixing bacteria. These microorganisms have enzymes called nitrogenases that can convert atmospheric nitrogen, electrons, and protons into ammonia, on their active site containing the MoFeS₇ cluster [34]. These enzymes have been studied to gain insights into their catalytic mechanisms and explore ways to incorporate them into artificial systems for sustainable ammonia production. Harnessing biological nitrogen fixation could offer a greener and more efficient alternative to conventional ammonia synthesis methods. Drawing inspiration from biological processes, researchers have suggested methods such as electrocatalytic and photocatalytic ammonia production under ambient conditions [35].

Generation 3 ammonia production method is to directly synthesize green ammonia by electrochemical production [36,37], using renewable electrical energy to convert nitrogen from air to ammonia in the presence of water under ambient conditions. In order to break the strong nitrogen bond, it needs a catalyst with high activity. Also, the catalyst should have a high selectivity to overcome the challenge of competing reactions of hydrogen evolution instead of incorporating hydrogen into ammonia, as well as a long lifespan, low overpotential, and high faradaic efficiency. This method still suffers from a low reaction rate, and the results reported in the literature are biased due to the difficulty in distinguishing the true activation of nitrogen from oxidized nitrogen impurities, which can lead to "false positives" [38,39]. This approach may not achieve the necessary scale and cost competitiveness compared to conventional or green Haber-Bosch processes for several decades. Nevertheless, if high efficiency could be achieved at potentials close to the thermodynamic reaction potential (i.e., low overpotential) it could eventually be the most cost-effective option as it eliminates the need for electrolyzers, air separation units, and the high-pressure high-temperature Haber-Bosch process [40]. One promising technology is the Li-mediated electrochemical production of ammonia in non-aqueous media, which could be considered practical [41–44] if it reaches sufficient activity. Unlike aqueous media, the non-aqueous media does not lead to the formation of Li(OH) and also helps with the stabilization of ammonia.

The electrocatalytic conversion of nitrate (NO₃⁻), one of the most abundant water source pollutants, to ammonia is also gaining increasing attention [45,46] as this process does not require breaking the strong N≡N bond to produce ammonia, while simultaneously being a promising water treatment strategy [47]. Most of the research is currently focused on developing more selective catalysts to direct the reaction towards ammonia production, among various other pathways [48].

Although advancements have been made, transitioning from the laboratory to practical application continues to be highly challenging [49]. Collecting nitrate and nitrite from wastewater is energy-intensive and yields limited amounts, making it impractical for industrial applications.

An extensive collection of review reports on recent progress and challenges in the electrochemical reduction of nitrogen and nitrogen oxides for achieving low-temperature operation have already been published in recent years discussing new electrode materials, electrolyte media, and reactor configurations [36,37,50–74]. Nonetheless, the ammonia yield is still low [47].

Other avenues are also being explored for sustainable ammonia synthesis. Photochemical and photocatalytic processes utilize light energy to drive the conversion of nitrogen and hydrogen into ammonia in ambient conditions [75]. These processes often involve the use of semiconductor materials as photocatalysts that can absorb light and initiate the necessary reactions. By harnessing solar energy, these approaches offer a renewable and sustainable route for ammonia synthesis. However, challenges remain in optimizing the efficiency and stability of photocatalysts and improving the overall reaction rates, as the catalyst activities are still three orders of magnitude less than the electrochemical synthesis method [28].

Plasma-assisted ammonia synthesis [76,77] involves the use of electrically excited gases or plasmas to activate the nitrogen molecules, facilitating their reaction with hydrogen to form ammonia. Plasma reactors operate at lower temperatures compared to traditional Haber-Bosch reactors and have shown potential for efficient and environmentally friendly ammonia production, but further research is needed to overcome the two main challenges of high energy input requirements for nitrogen disassociation and an unwanted reaction path toward ammonia decomposition [78].

Recently, mechanochemical ammonia production in a ball-milling reactor has emerged as a promising method. This process allows ammonia synthesis to take place under mild conditions, 45 °C and 1 bar pressure, in the presence of iron powders, resulting in a final ammonia concentration as high as 82.5 vol % [79]. The collisions within the reactor contribute to the creation of defect sites, and facilitate the dissociation of N₂, while the impacts also aid in the desorption of strongly adsorbed intermediates. The slow reaction rate in the milling batch reactor needs to be improved to match the traditional Haber-Bosch process [80].

An alternative thermocatalytic process to the Haber-Bosch is the chemical looping ammonia production (CLAP) which utilizes nitrogen carriers to react nitrogen and hydrogen at atmospheric pressure [81]. When concentrated solar power is used to provide the thermal energy for the reaction [82], and hydrogen is produced from renewable resources, it has the potential to be completely carbon neutral. In chemical looping, a reaction breaks down into sub-reactions, using intermediates that are regenerated along the way [83]. CLAP involves nitrogen activation by generating metal nitrides [84,85] or metal imides [86], ammonia production by reacting nitrogen carrier materials with water or hydrogen, and finally regeneration of intermediate materials [81]. Ammonia generation through the hydrolysis of metal nitrides leads to the creation of stable oxides or hydroxides. As a result, the endothermic regeneration of metals or metal nitrates demands elevated temperatures. For instance, the regeneration of ALN from Al₂O₃ necessitates temperatures as high as 1500 °C [87]. The other pathway is the two-step process of metal nitration and exothermic de-nitration through hydrogenation [86]. Different nitrogen carriers have been investigated for their kinetics and thermodynamics of nitrogen fixation and ammonia evolution [84,85,88]. However, The widespread application of CLAP has been hindered by the necessity for elevated temperature [85] and having relatively low ammonia yields [88]. The major challenges are the nitrogen migration process and slow reaction on the surface of solid materials [89].

Continued efforts have been made to demonstrate the promises of various non-conventional decarbonized ammonia synthesis routes by activation of nitrogen [24] via electrocatalytic [36,37], enzymatic [90],

photocatalytic [91,92], plasma-catalytic [25,93], and photo-electrocatalytic [94] processes. However, the pursuit of materializing their practical implementation is hindered by the fact that these process technologies have not successfully managed to achieve sufficiently high ammonia yield rates. Therefore, the thermocatalytic pathway via the Haber–Bosch process, which is the conventional ammonia synthesis process, has remained as the most prevalent method of production, and by extension the most prevalent field of research and developments in decarbonizing industrial ammonia [24].

In the Haber–Bosch process, the exothermic reaction of hydrogen and nitrogen is favored by low temperatures and high pressures. However, on the Fe-based catalysts used in the industry, higher temperatures are required to dissociate N_2 , which is the rate-limiting step of the ammonia synthesis reaction. To enhance the single-pass conversion of ammonia and facilitate its liquefaction and separation from the unreacted hydrogen and nitrogen gases, an increased pressure is necessary [22]. The presence of ammonia decelerates the synthesis process as the reaction advances, in accordance with Le Chatelier's principle. Consequently, it is imperative to continuously remove the ammonia from the product stream containing the unreacted gases to minimize the limitations imposed by the reversible reaction [23,95]. The requirement for such elevated temperatures and pressures in the Haber–Bosch process can only be materialized by high capital and operating costs, which will only be feasible in reasonably large-scale and centralized ammonia manufacturing plants [22]. In view of these limitations, this review is undertaken to summarize recent research and advancements made to the loop and its key components (i.e., the reactor and the catalyst, separation, and purge gas recovery technologies), and analyse the level of effectiveness and success of their implementations.

3. Advances in Haber–Bosch process

After acquiring Haber's patent, BASF initiated industrial-scale ammonia production in Oppau, Germany [96]. Since then, in licensing new technologies, five key players have been at the forefront: KBR (Kellogg Brown & Root), Topsoe, Casale, Thyssenkrupp, and Linde plc. Advancements in ammonia synthesis have led to larger plant capacities, enhanced reliability, increased efficiency, and reduced capital costs.

3.1. Ammonia synthesis loop

The above improvements mainly emerged in two phases: initially with the launch of large “single-train” (single-stream) ammonia plants and subsequently through efforts to optimize plant efficiency. A significant milestone was achieved by M. W. Kellogg by integrating various components of the plant, paving the way for the development of a more comprehensive single-train ammonia plant in the 1960s. Single-train in ammonia production involves conducting the entire process, from hydrogen production and nitrogen extraction to the Haber–Bosch synthesis of ammonia, in a continuous, integrated flow. This resulted in enhancing efficiency through economies of scale and integrated design. This configuration allows for the seamless flow of materials and energy, reducing operational complexity and capital costs. The integrated nature of these plants facilitates energy recovery and reuse within the process somewhere else, further improving efficiency. The energy efficiency of contemporary ammonia production facilities is enhanced through advanced heat and cold recovery systems [97]. In the majority of these plants, the heat generated from the ammonia synthesis reaction is employed typically for preheating boiler feed water. The energy requirements of contemporary ammonia processes have been reduced from 45 MJ/kg to 28–30 MJ/kg ammonia [98]. The minimum theoretical energy requirement is the HHV of ammonia itself, approximately 22.48 MJ/kg for gaseous and 21.32 MJ/kg for liquid ammonia at 25°C [27].

Another significant improvement involves using centrifugal compressors, particularly in compressing synthesis gas. While reciprocating

compressors were commonly used until the early 1960s, contemporary methods predominantly involve the use of centrifugal compressors, except in a few small-scale plants [27]. Centrifugal compressors, operate at high speeds and can be driven by high-pressure steam turbines. The process's excess heat is harnessed to produce high-pressure steam. This steam is then utilized to operate the centrifugal compressors. This change brought about higher production capacities, and improved reliability due to less frequent maintenance needs. Additionally, the majority of the water is typically eliminated through condensation during compression stages, reducing its concentration to a very low level. Also, the use of centrifugal compressors in modern methods has greatly decreased the transfer of oil into the synthesis loop [27]. Subsequent advancements primarily focused on hydrogen production.

Advancements over the years in syngas production, including the introduction of low-temperature shift and methanation catalysts, have significantly improved gas quality by reducing the compounds with oxygen. Apart from improvements in the plant design and upstream syngas production via SMR and utilization of centrifugal compressors, other improvements have been achieved through optimization in the catalyst, reactor, and separation process designs, which will be discussed in the following sections.

Due to thermodynamic constraints, only a fraction of the hydrogen and nitrogen can be converted into ammonia in a single pass. To enhance efficiency, the unreacted hydrogen and nitrogen are recycled back into the process after the ammonia has been extracted, thus forming a continuous ammonia synthesis loop. The recycling of the unreacted hydrogen and nitrogen is necessary, as typically the conversion per pass in the ammonia reactor ranges from 12 % to 16 %; more advanced plants have achieved conversion rates of up to 19 % [99]. In a typical ammonia synthesis loop, as depicted in a simplified form in Fig. 2, compressed hydrogen and nitrogen are fed into one or more heterogeneous catalytic reactors, and then a heat exchanger is utilized to cool down the product gas and recover the heat of the exothermic reaction generally for boiler water heating and reactor feed pre-heating. A separator system retrieves the ammonia product and recycles unreacted hydrogen and nitrogen. A circulator (recycle compressor) moves the gas and pressurizes the hydrogen and nitrogen in the loop to compensate for the lost pressure in the loop. The power needed for circulation is relatively minor compared to the power necessary to compress the fresh feed, owing to the low-pressure differential in the synthesis loop.

There are different types of synthesis loop configurations used in the production of ammonia. One layout is shown in Fig. 2, which has the recovered and make-up gas addition points and the withdrawal of the purge gas stream after the ammonia separator. This aligns with the optimal approach which is to separate ammonia before being diluted with the make-up gas, withdrawal of the purge at the point with the highest inert concentration, lowest system pressure, and lowest ammonia content [99]. Also, this leads to a decrease in energy consumption due to the compression of the least possible amount of gas. However, the possible disadvantage of this layout is that all impurities present in the make-up gas are passed directly to the ammonia reactor. Therefore, this layout is used mainly in cases where catalyst poisons are removed very effectively from the feed gas with methods such as molecular sieve dehydration or liquid nitrogen wash [100]. Alternative synthesis loops involve withdrawing the purge gas and adding make-up gas upstream of a separator or between two separators [101]. In loops operating at high pressure (above about 250 bar), it has been found advantageous to install two ammonia separators [100].

Exergy analysis is a valuable tool for identifying and quantifying sources of inefficiency and opportunities for improvement in complex industrial processes [97]. Penkuhn and Tsatsaronis [102] have examined two different ammonia synthesis loop configurations of a three-staged adiabatic reactor with intermediate quench cooling and a reactor designed with multi-tube indirect cooling, utilizing a cooling medium, to identify the real thermodynamic inefficiencies and possible improvements in the components of the loop. They concluded that

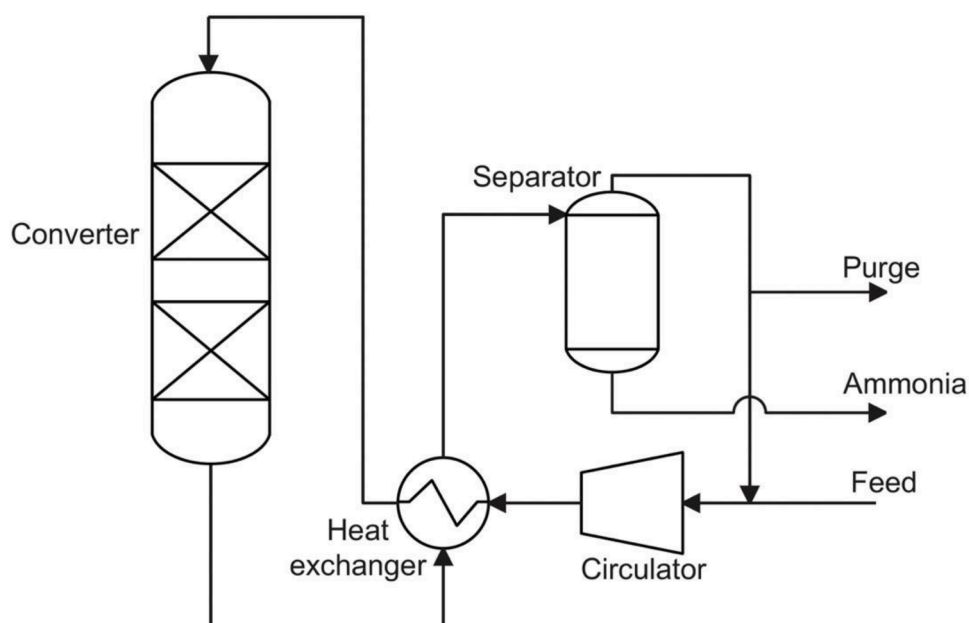


Fig. 2. Ammonia synthesis loop in a simple form.

significant improvement opportunities lie in enhancing the waste-heat-steam-generator and the main compressor, and while the improvement potential in the reactor design is limited, its critical role in heat integration makes it a key component.

3.2. Ammonia synthesis reactors and catalysts

3.2.1. Reactor design

The ammonia reactors are mainly comprised of a pressure vessel that accommodates cartridge sections for catalyst load and a heat exchanger

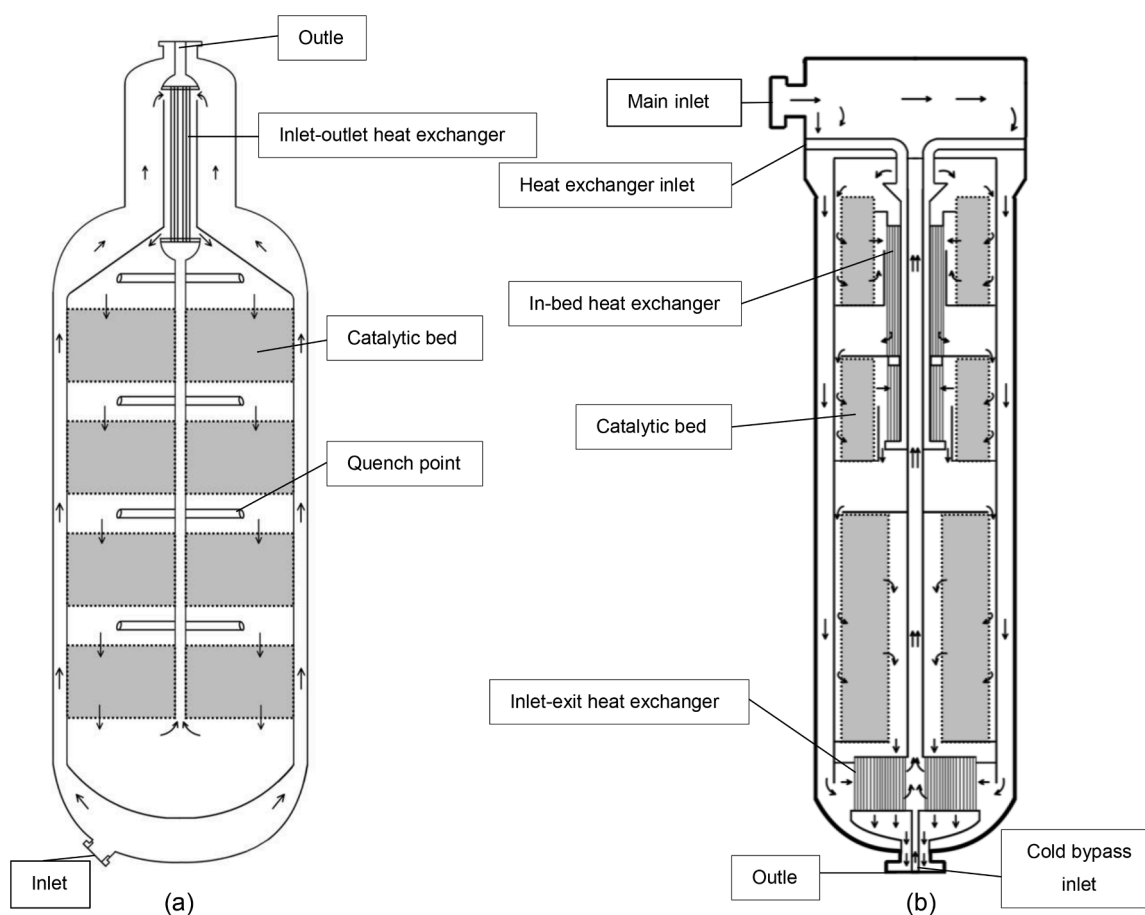


Fig. 3. Four beds Kellogg quench reactor (a), and TOPSOE's S-300 reactor (b).

(s). The cartridge is commonly made of stainless steel, and it can be moved out from the pressure vessel easily. Each cartridge section can be categorized based on the syngas flow direction and heat management and recovery systems. The flow types include axial, radial, and axial-radial, while heat management options encompass quench-cooling or indirect-cooling using heat exchangers.

In early design examples, the reactor is operated using an axial flow configuration. A high gas flow rate moving axially through a catalyst bed in such a vessel results in a high pressure drop, leading to increased circulator power consumption. However, the introduction of a radial flow reactor was found to effectively reduce the pressure drop across the catalyst bed. This design allows for the utilization of smaller particle catalysts and minimizes diffusion limitations. The feed gas is introduced into an annular space that surrounds the beds. As a result, the gas moves radially through the bed, traveling a short distance but with a larger cross-sectional area, until it reaches a central exit-gas collector.

Because ammonia synthesis is exothermic, maintaining an effective temperature control is a fundamental aspect of reactor design, as higher temperatures lead to lower equilibrium ammonia concentrations. Two methods of temperature control and heat recovery are available. A quench-cooled reactor uses a cooling system by injecting cold feed gas between separate catalyst beds. In this method, an internal heat exchanger also can be utilized to preheat the feed gas with the recovered heat from the products. The second method involves indirectly cooling the catalyst bed using heat exchangers with a cooling media of water for raising steam or the feed gas for pre-heating. This method results in a higher ammonia yield per pass because the ammonia concentration in the reactor is not reduced by blending with the quench gas.

The M. W. Kellogg's 3 or 4-bed quench reactor is a prominent example (Fig. 3a) of a quench-cooled reactor. The syngas is divided into mainstream and multiple quench flows. Quench gas distributors are located between the beds. The mainstream gas passes through a heat exchanger to be pre-heated by the product gas and then enters the first bed. The output gas from each bed is quenched and diluted with fresh feed before entering the next bed and eventually exits after passing from the other side of the heat exchanger.

For indirect cooling with heat exchangers, the cooling medium can be either the boiler feed or process feed gas. Heat exchange with the catalyst bed can occur intermittently or continuously using cooling tubes within the bed. Temperature control at intervals divides the catalyst bed into adiabatic beds. An example of a modern ammonia reactor with indirect cooling and radial flow is TOPSOE's S-300 ammonia reactor (Fig. 3b).

The requirement for extreme conditions in the exothermic reaction of nitrogen and hydrogen in the Haber-Bosch process can be attributed to the activity of the catalysts that necessitates temperatures exceeding 200 °C [103,104]. To achieve sufficient activity, the kinetic performance of the ammonia commercial catalysts typically requires temperatures of 400–600 °C. To reconcile this conflict, the process employs high pressure, typically ranging from 100 to 300 bar [24,104–106]. The harsh conditions of high pressure and temperature within the vessel could worsen the risk of hydrogen embrittlement and nitriding. Specific materials are needed for the reactor components depending on the operational conditions. To reduce capital costs, the utilization of cold-wall configurations is common when a high-temperature catalyst bed is housed within an internal cartridge. To maintain a lower temperature in the pressure vessel exposed to a higher-pressure differential, a flow of cold syngas is directed between the cartridge and the vessel, as shown in (a) and (b) of Fig. 3. For pressure vessels, the use of thick carbon or low alloy steel sheets has been reported, while high alloy steel type 304 is used for catalyst cartridges and heat exchangers [107]. Incoloy and Inconel are also employed for components exposed to higher temperatures.

There are several reactor options offered by different companies. Casale offers an axial-radial flow reactor. Their axial-radial technology (Fig. 4) optimizes catalyst utilization by efficiently using the entire

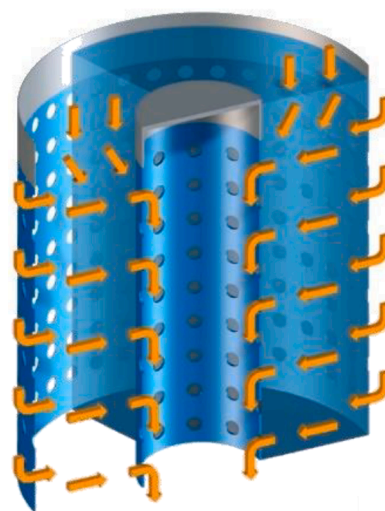


Fig. 4. Casale axial-radial catalyst bed (re-printed from Casale [108]).

catalyst volume while minimizing pressure drop during gas flow through the reactor. In this design, while radial flow results in less pressure drop, the syngas is also processed in an axial flow downward through the reactor. This reactor is utilized in a wide range of ammonia processing plants ranging from 50 t/d to 6000 t/d ammonia production [108].

To enhance the conversion rate with a large catalyst volume, minimize pressure drop, and optimize the utilization of reaction heat for the production of high-pressure steam, Uhde (part of Thyssenkrupp) provides three distinct radial reactors [109], each featuring one to three beds that can be arranged within one or two reactor configurations.

In a similar manner, TOPSOE offers reactors up to three catalyst beds. The TOPSOE S-50 reactor [110] is a single-bed radial-flow reactor added downstream of the main reactor in series to increase ammonia conversion. It comes in two versions: hot and cold shell. The hot shell version is used when the upstream reactor is designed without an internal inlet-exit heat exchanger. The heat generated in the initial reactor guarantees that the input to the S-50 reactor is adequately heated for the ammonia synthesis process. TOPSOE's S-300 [111], featuring radial flow and indirect heat exchange, represents the company's most recent reactor. It offers three catalyst beds in comparison to the older two-bed S-200 model [112]. The S-300 basket can also be installed within an existing reactor pressure shell to revamp an older reactor.

It has been proposed that magnetic heating, applied to catalysis, is a technology capable of shifting various catalytic reactions from being limited by reactor heat transfer to being governed by process kinetics [113]. Given its added benefits like rapid start-up and compatibility with renewable electricity, its application in ammonia synthesis is being anticipated [114]. HØjlund Nielsen Poul [115] patented an energy-efficient method to start up heating the catalyst using inductive heating. The catalyst can be ferromagnetic, antiferromagnetic, or non-magnetic, and in the case of a non-magnetic catalyst, it can be mixed with a ferromagnetic material to enhance the inductive heating process. Hu et al. [116] describe heterogeneous catalysts suitable for ammonia synthesis under microwave irradiation heating, along with methods for their preparation and the ammonia synthesis process itself.

In a few studies, the optimization and modeling of industrial ammonia synthesis reactors have been carried out. A study [117] developed steady-state models for a vertical four-bed axial flow ammonia reactor. After verifying the model with data from Notore Chemical Industry, the effects of inlet flow rate, pressure, and temperature on the reactor's performance were calculated. Another research [118] focused on a three-bed vertical reactor with radial-flow catalyst beds, employing Aspen HYSYS and a modified form of Temkin and

Pyzhev kinetic equation, leading to optimized reactor temperature profiles and potentially to increased ammonia production. Akpa and Raphael [119] proposed a one-dimensional simulation and optimization scheme for optimizing the inlet temperatures of an ammonia synthesis reactor. In another study, a steady-state one-dimensional model for an axial-flow reactor was developed [120] to determine the optimal design parameters based on the data collected from the industry.

3.2.2. Catalyst

A highly active ammonia synthesis catalyst is an important enabler towards achieving effective performance by process intensification of the Haber-Bosch process. Walking through its historical journey from the 1900s to 1950s, in essence, ammonia plants worldwide are mostly using Fe-based catalysts promoted by metal oxides such as Al_2O_3 and K_2O . Magnetite iron (Fe_3O_4) catalyst is commonly used in the Haber-Bosch process to produce ammonia from nitrogen and hydrogen in the ratio of 1:3 under harsh reaction conditions. Typically, elevated temperatures ranging from 400 to 600°C are required to achieve sufficient catalytic activity for the dissociation of the triple bond in the dinitrogen molecule [104]. This molecule is very stable, with a high bond dissociation energy of 941 kJ/mol.

The cost of catalysts has a minimal impact on the overall production cost, accounting for 0.26 % – 0.6 % of the total cost per ton of ammonia [121]. This cost factor had called for the use of more active but rare materials. As a result, Ru-based catalysts, which had shown activity of 10 folds greater than that achieved by Fe but at low pressures, came into commercialization after early 1990s [104,122]. However, their adoption has been limited with a record of only 16 plants around the globe using them after 20 years of debut for being less attractive due to their high cost [24]. In addition, Ru-based catalysts on carbon carriers are prone to the risk of carbon material methanation during ammonia synthesis. This process depletes the active carbon carrier and shortens the catalyst's lifespan [122]. Another issue reported in the literature is the hydrogen poisoning of Ru-based catalysts, which reduces their lifespan. This however has been overcome by an electride, $\text{C}_{12}\text{A}_7\cdot\text{e}^-$, which was proposed as an electron donor for a Ru-based catalyst [123]. This electride demonstrates high electron-donation and chemical stability, and reduces hydrogen poisoning due to its reversible hydrogen absorption/desorption capability. Furthermore, although Ru-based catalysts exhibit significantly higher activity compared to Fe-based catalysts, it requires a low-pressure operation [24]. Consequently, condensation, which takes place at moderate pressures becomes challenging. There is a non-linear relationship between pressure and compressor energy. Therefore, conventional high-pressure systems appear more attractive since there is no requirement for additional compression during the ammonia separation step [122], which goes back to square one showing no considerably significant effect of using the expensive Ru. Therefore, Fe-based catalysts continue to lead in the ammonia synthesis catalyst market. The identification of wustite iron (FeO) as a substitute for magnetite iron enabled a reduction in pressure to 100 bar [124].

Gerhard Ertl's notable contributions to understanding the mechanism of the ammonia synthesis reaction were recognized with the Nobel Prize in Chemistry in 2007. As suggested by the mechanistic steps of the process on the catalytic active sites, the rate of ammonia production is limited by the nitrogen dissociation step due to its dependence on the energetic level of the nitrogen species on the catalyst surface [125], which was determined experimentally for the first time by Emmett and Brunauer [126]. A few years later through surface science studies, the observations of reaction intermediates [127] became possible. Theoretically, the transition states of key reaction intermediates and elementary steps were not fully understood in the measurements of the catalytic activity towards nitrogen dissociation until the 21st century when theoretical calculations using density functional theory were introduced [128]. The Haber-Bosch process is kinetically complex, involving the dissociative adsorption of N_2 and its reduction by H_2 , with

the formation of intermediate NH_x species that ultimately leads to the production of ammonia.

To develop a highly effective catalyst, a metal with a sufficient electronic density near the Fermi level for back-donation is desirable. Low work functions, which are seen in metals like sodium and potassium, are also desirable, as they allow the metal to give off electrons easily. However, these metals are unstable and are therefore used as promoters rather than as the main catalyst. The Sabatier principle was used in the development of an ammonia synthesis catalyst, leading to the derivation of volcano plots, which indicate the relationship between the catalytic activity of the metal surface and the strength of the interaction of the N_2 molecule with the surface. The volcano plots serve as a crucial starting point in the steps involved in developing more sophisticated catalysts as they provide the indication of whether the selected metals would achieve a sufficient catalytic activity or if the reaction is limited by either desorption or adsorption steps of the nitrogen species on the metal surfaces that would result in turnover frequency to be too low [129]. The ammonia synthesis catalyst should have two different active sites for hydrogen and nitrogen, to prevent competition between the two species. Additionally, too weak, or too strong of an interaction between the catalyst and N_2 can result in limitations in the desorption of the products and adsorption of the reactants. A moderate level of interaction, known as the Sabatier optimum [130], is considered desirable. These descriptors have successfully been used to formulate low-temperature ammonia synthesis catalysts that are of faster rates than the Fe-based, such as the bimetallic [131] and metal nitrides [132] catalysts. For example, a patent by Natali et al. [133] details a method for ammonia synthesis using rare earth (mostly La-based) nitrides at room temperature and low pressure, which also optimizes the process by using a vacuum or inert atmosphere to facilitate the reaction of hydrogen with nitrogen released from the rare earth material to produce ammonia. Additionally, incorporating a second catalytic site could disrupt the correlation between adsorption and transition-state energies on transition metals, thereby enabling more efficient catalysis. Hydride in LiH removes activated nitrogen from transition metal sites [134] and provides hydrogen to form LiNH_2 , which eventually produces NH_3 and regenerates LiH.

The quest for metal-based catalysts to facilitate the reduction of dinitrogen has been a subject of intensive research in catalysis for over a century to develop catalysts that can sustain production rates at temperatures below 400°C. Despite significant scientific progresses made in the catalysis research area, the commercial catalysts employed in industrial ammonia synthesis today bear a striking resemblance to those developed in the early to mid-1900s [24], with Fe- and Ru-based catalysts being the common ones. Indeed, this area has been carefully studied over the last century but only demonstrated small improvements at a large scale. Therefore, in this review, we are not focusing on the recent advancements in ammonia catalyst development. A wide range of works in the search for metal-based catalysts for dinitrogen reduction in ammonia synthesis reported since the past 100 years have been compiled and published in recent literature as review articles, summarizing their preparation techniques using various active materials and precursors, effects of promoters, and their activity performances [24, 104,124,135–137].

3.3. Ammonia separation technologies

The commercial Haber-Bosch process is operated in a conventional reaction-condensation synthesis loop. After its synthesis reaction over a conventional catalyst at a pressure of 100–300 bar and temperature of 400–600°C, condensation then takes place at –20°C in an high-pressure separator to separate the ammonia product from the unreacted gases, where the liquefied ammonia is continuously flashed [138–140]. This configuration does not only require the integration of heat exchangers and ancillary equipment of huge investments [23], but the cryogenic conditions requirement for ammonia separation is also energy intensive

due to the high-pressure phase change [141]. To lead towards a more efficient separation at reduced pressures, the Haber-Bosch synthesis loop needs to be reconfigured. An ideal strategy that has recently received attention is the development of non-cryogenic ammonia separation technologies to effectively operate ammonia manufacturing at low pressure, leading to order-of-magnitude energy savings [22].

One way to reconfigure the Haber-Bosch synthesis loop is to replace the ammonia condensation with a sorption-enhanced process by using ammonia-selective materials such as adsorbents [142] and absorbents [95,141,143]. Excellent ammonia selectivity over hydrogen and nitrogen allows the sorbent to be regenerated through pressure and/or temperature swings to release the concentrated ammonia product so that a more efficient ammonia separation and an overall low-pressure operation can be achieved [22]. This is visualized in Fig. 5.

Although the adsorption and absorption processes conceptually are both based on the materials' selectivity towards ammonia, they should be distinguished by their working mechanisms and principles, having differences in their capacity and capability performance. Adsorption is a bonding between adsorbate molecules and a solid surface either physically or chemically [23]. The binding strength of NH_3 , N_2 , and H_2 molecules to the surface is weak and is not sustained at high temperatures [141]. A general comparison between adsorbent materials and absorbent materials is that the former have a lower binding energy and a much lower potential capacity due to surface area limitation, and therefore are inherently less selective [23]. In the development of adsorbent materials, not only the adsorbent particles must be selectively specific towards ammonia, but the challenge also lies in making the adsorption capacity high towards ammonia at high temperatures. Typically, an adsorption material such as Cu-Y zeolite is used in this application, with the equilibrium concentration of NH_3 on it can be calculated using the Langmuir isotherm, where the equilibrium amount of adsorbed NH_3 decreases with increase in temperature [142].

Absorbents, on the other hand, do not only offer a higher capacity and ammonia selectivity since they selectively incorporate molecules into the crystal lattice, but also have higher stability at high temperatures [23]. Absorption, which relies on the surface gas–solid reaction and bulk diffusion within the solid absorbents, would enable close to 100 % ammonia selectivity over other gases, unlike the adsorption that is merely a surface-based phenomenon [95]. The absorption process occurs by ammonia entering the crystalline matrix of the absorbent, where the ammonia selectivity derives from the persistent stability of the metal amine complexes at temperatures of as high as 400°C [141]. In the development of the absorbent materials, the ammonia absorption, which is the take up of ammonia by the salts, is not effective if the ammonia molecules are only being retained on the crystal surface. To make it effective, we need to develop a material that can absorb the

ammonia molecules that go into the crystal salts [143]. Using MgCl_2 as an example of the ammonia-selective absorbent, Fig. 6 shows the schematic of ammonia being selectively absorbed into the MgCl_2 crystal lattice [23].

Although various adsorptive ammonia separation works have been demonstrated and published in recent years, they might not be as worthy as the absorptive separation for further potential exploration in ammonia separation technologies due to their lower effectiveness, based on general assessment. Therefore, our current work will not take a closer look into this technology as part of our review. Nonetheless, some selected examples of adsorptive separation work reported in the literature are cited here for the benefit of readers who might want to investigate this technology further. Starfire Energy patented an ammonia synthesis process via modification of the Haber-Bosch process using a monolithic catalyst in the ammonia reactor to reduce the pressure drop and multiple adsorbents for the ammonia recovery over zeolite as the adsorbents. In their patent, the nitrogen was supplied from a PSA unit, while the hydrogen was produced by water electrolysis [145]. A more detailed study with regards to the choice of various zeolite adsorbents performed by Matito-Martos et al. [146] using Monte Carlo simulations indicated that the removal of ammonia from the ammonia synthesis reaction by adsorption was successful in not only shifting the reaction equilibrium but also in increasing the ammonia yield rate at lower pressures. A study by Lin et al. [147] that attempted to concentrate the dilute ammonia stream via rapid PSA at low temperatures of as low as 25°C revealed that although the use of the rapid PSA-assisted adsorber enabled recovery of >95 % ammonia from the reaction effluent gas, the adsorption relied on a weak van der Waals bond that is ineffective under the reaction conditions where the temperatures are high.

Using absorption, not only ammonia is potentially separated from the reactor effluent completely [22], but the technology employment also enables operation at pressures 10–20 times lower than the conventional high-pressure ammonia manufacturing, while still producing comparable ammonia rates [95]. The absorption technology in the ammonia loop is schematically shown in Fig. 7. It requires two absorption units with alternating states of absorption and desorption. The desorption process can be approached in two ways. One method involves increasing the temperature while keeping the pressure high. The other method involves reducing the pressure to atmospheric levels without raising the temperature, which saves energy by eliminating the need for heating, but this approach requires ammonia compression before condensation [139].

A kinetic investigation of ammonia absorption on MgCl_2 at varying pressures and temperatures by Huberty et al. [149] suggested that the absorption process is diffusion-limited. Therefore, the key to success in the ammonia absorption process relies on the development of

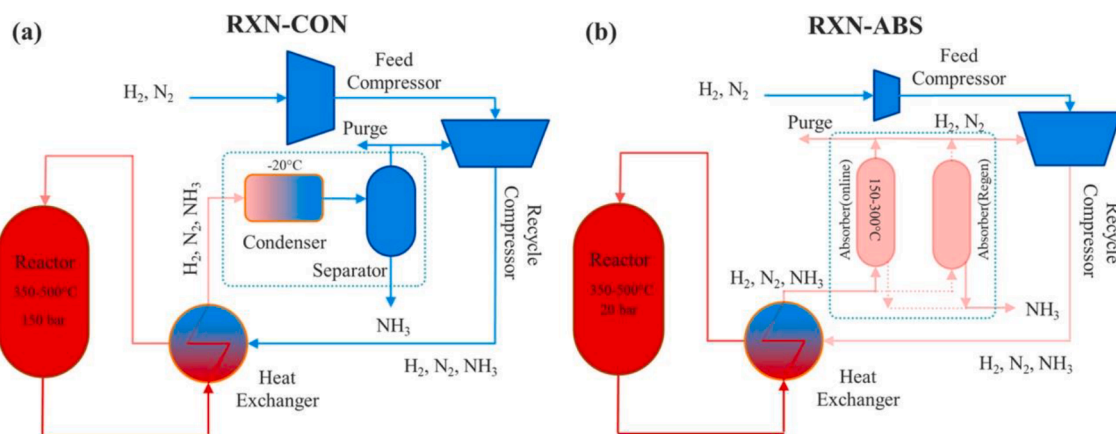


Fig. 5. Ammonia synthesis loop schematic block diagrams based on: (a) conventional reaction-condensation, and (b) reaction-adsorption/absorption. Reprinted with permission from [144]. Copyright 2016 American Chemical Society.

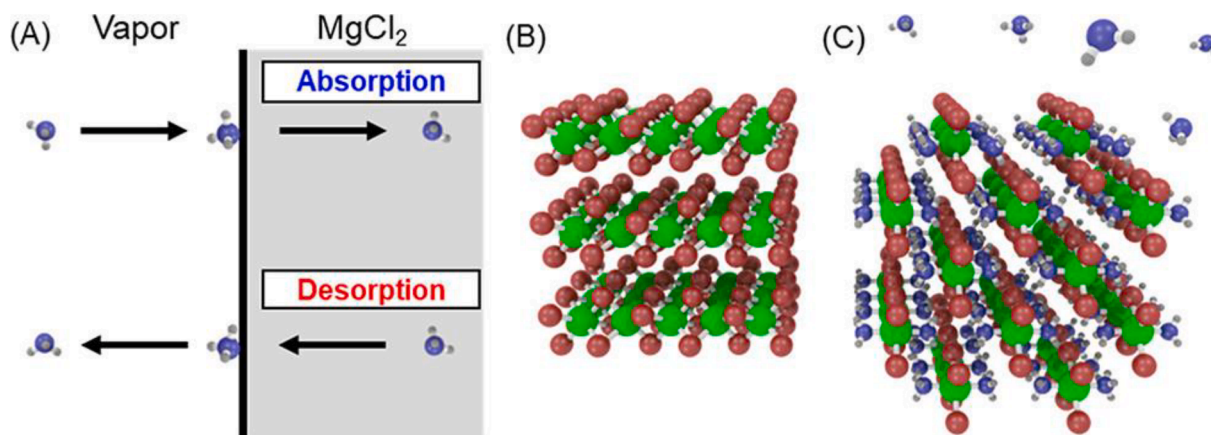


Fig. 6. Schematic description of ammonia absorption and desorption within MgCl_2 salt. (A) Ammonia vapor is adsorbed onto the salt surface and absorbed into salt crystal lattice. Ammonia is desorbed within the salt by diffusion to the surface, leaving into the bulk vapor phase. (B) Crystal structure of MgCl_2 without ammonia molecules. (C) Crystal structure of MgCl_2 with ammonia molecules [23]. Reproduced with permission from American Chemical Society, Copyright 2020.

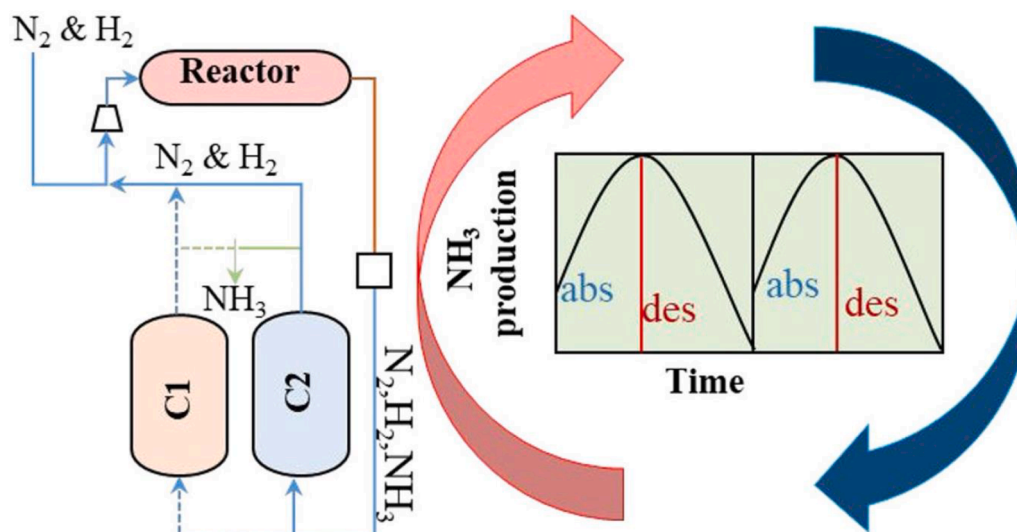


Fig. 7. Schematic of absorption technology in ammonia loop [148]. Reproduced with permission from American Chemical Society, Copyright 2022.

absorbents that can take up and release ammonia efficiently while being stable under the operating conditions. Therefore, efforts should be made to design absorbents with high working capacity that can sustain rounds of reproducible cycles [22]. The use of absorption into crystalline salts (i.e., metal halides) has been offered [143] to address the issue of low partial pressure associated with ammonia synthesis catalysts. Metal halides exhibit a high level of ammonia selectivity and absorption capacity and can isolate ammonia under relatively high temperatures (200–300°C) and very low partial pressures (0.002–0.1 bar) [3,150], which give a significant advantage for use in the Haber-Bosch ammonia synthesis loop [148], and especially a low-pressure loop.

In a study by Malmali et al. [143], it was demonstrated that achieving the equilibrium capacity of ammonia absorption into various unsupported metal halides in breakthrough experiments was difficult, typically showing less than 1–5 % of their thermodynamic capacity in general. Indeed, not only was the thermodynamic capacity low, but the breakthroughs were unreproducible, and the capacity decreased after every cycle. Therefore, supported metal halides have been explored as the absorptive materials to achieve breakthrough reproducibility cycles [22].

High reaction temperatures have a minimal influence on the overall production rate at the optimum recycling rate since the reaction-absorption system is controlled by the equilibrium conditions

when the reaction temperature reaches a certain point. Pressure on the other hand has a significant impact on the reaction-absorption system, where higher pressures are more favourable, but this deviated from the objective of achieving low processing pressure operation. To sustain high production rates at lower pressures, the reaction temperatures need to be lower. The only way to achieve the low-temperature operation is by using highly active catalysts that give faster synthesis rates [148]. This appears to have not been able to resolve the objective of achieving low-pressure operation by engineering design modification without replacing the utilization of the existing commercial Fe-based catalysts.

On the ammonia separation side, although absorption operation at higher temperatures can reduce the required heat exchange in the Haber-Bosch synthesis loop, it comes with the price of incomplete separation due to reduced absorption equilibrium capacity. Since the partial pressure of ammonia is the absorption rate-limiting step and absorption at high temperature is desired, the reaction conditions should therefore be selected accordingly so that a higher single-pass conversion can be achieved [148]. We continue with reviewing some selected process optimization works pertaining to ammonia separation via absorption which attempted to achieve optimal conditions at the maximum attainable ammonia production rate.

Based on the simulation by Palys et al. [151], the most energy-intensive aspect is the heating required to raise the temperature

of the absorbent by 300°C for regeneration. Nonetheless, this heat can be obtained from the exothermic ammonia synthesis reaction, which can reduce the overall cost, compared to the conventional ammonia separation by condensation. In reality, however, the regeneration of the absorbent requires a lower temperature compared to the simulated 300°C, and therefore, the energy cost would become as competitive as the conventional condensation process [3].

Using MgCl_2 absorbents supported on SiO_2 , Kale et al. [23] optimized the ammonia cyclic uptake and release by studying the effects of absorption and desorption temperature and pressure, and desorption time on the absorbent's non-equilibrium working capacity. The optimization aimed at achieving absorption temperatures and pressures that use minimal energy input, while maximizing the ammonia uptake and release within a reasonable time. First, the ammonia was allowed to break through. Over this time, the temperature profile showed evidence of the heat of absorption of ammonia, indicated by the largest temperature increase right after the absorption began. Then, the bed was heated for regeneration. Heating the bed was hard because the thermal conductivity of the bed was low, and the thermal diffusivity of the bed was also low due to the high heat capacity of the bed per volume of bed. The desorption kinetics were limited by the initial heating rate. The desorption step showed two regions of ammonia release: (I) a fast, near-instantaneous release, and (II) a very slow release; exhibiting a characteristic rate of desorption controlled by ammonia diffusion. The desorption was the slowest step in the cyclic process. In the final step, the bed was cooled, whereby again, the thermal diffusivity was the key factor, dependent on the thermal conductivity of ammonia and the heat capacity of the absorbent. Each step included the manipulation of temperatures, flows, and pressures for a specific amount of time.

The findings revealed requirements for low and high temperatures for the absorption and desorption operations respectively to achieve high non-equilibrium working capacities. To maximize the amount of

ammonia being absorbed and desorbed per hour, the amount of time spent in regenerating the sorbents should be reduced to achieve fast desorption, while still maintaining a sufficiently long time to heat the sorbents. The above requirements were derived based on the findings that the regeneration is limited primarily by ammonia diffusion through the solid sorbents and into the bulk gas phase when longer desorption times and small ΔT (between the absorption and desorption) cycles were used [23].

The transient behavior of the reaction-absorption process in the ammonia synthesis loop is critical for strategizing a good process control and optimization. Typically, the research in this area has been limited to batch and short-term fed-batch tests, where the ammonia production rates were indirectly calculated based on pressure drop in a very short period of operation, or by simplified theoretical models. For better accuracy, the process should indeed be investigated at constant pressure and continuous mode of operation. With an aim to obtain more insights into the transient nature of the reaction-absorption process so that the continuous ammonia production can be maximized, Nowrin and Malmali [148] presented a systematic study to optimize the continuous reactive separation of ammonia using absorption into unsupported CaCl_2 salts by reporting the rates of continuous ammonia production at a fixed operating pressure in a fed-batch process, executed in three modes as illustrated in Fig. 8.

The first scheme was focused solely on the ammonia synthesis reactor, where the performance of the catalyst was studied as an effect of combinations of different reaction temperatures, pressures, and space velocities. In the second scheme, the effectiveness of the absorbent was tested using space velocity and a wide temperature range at different ammonia partial pressures as the key parameters in a single-pass reaction that was followed by absorption. Using the optimized reaction and absorption conditions evaluated in the two schemes, a series of multi-pass fed-batch reaction-absorption tests were conducted in the third

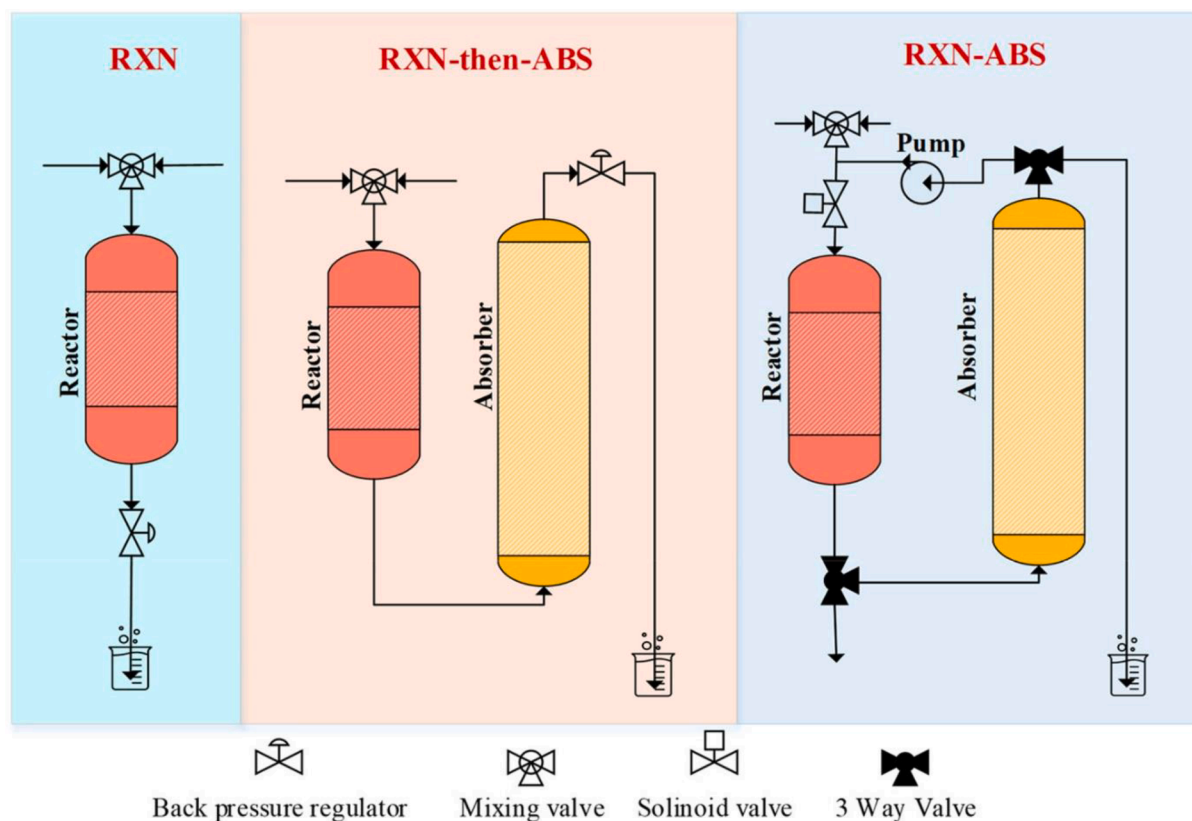


Fig. 8. Modes of ammonia production process in the optimization study by Nowrin and Malmali [148], without and with separation and recycling. Reproduced with permission from American Chemical Society, Copyright 2022.

scheme at a fixed pressure to gain insights into the transient behavior of the process. It was found that equilibrium limitation does exist, where the reaction temperature is the main parameter that controls the overall ammonia production rate regardless of the absorption temperature. For continuous operation, the findings revealed that the recycling rate influenced the production rate more greatly than reaction temperature and pressure, but the recycling rate must be selected carefully by considering the absorption temperature. Based on these findings, Nowrin and Malmali [148] proposed the use of low-temperature absorption and high-temperature desorption to achieve a higher ammonia production rate, which is very well aligned with the proposal put forward by Kale et al. [23] seen earlier. They also proposed that the dissipated heat from the reactor and absorber column should be recovered for utilization during desorption.

Liquid absorbent has also been investigated in ammonia absorption to establish an energy-saving system by finding low volatility absorbents with excellent ammonia absorption efficiency. Ionic liquids have emerged as outstanding absorbents for gases such as CO_2 , CH_4 , SO_2 , fluoromethane, and fluoroethane in gas separation and purification processes, and ammonia is not an exception. The use of ionic liquids has made obtaining high-purity liquid ammonia product possible due to their high thermal and chemical stability with designable structure, and low vapor pressure/volatility and low heat capacity and no latent heat of evaporation that avoid large amount of evaporation. Since the discovery of ionic liquids' potential in ammonia recovery by Yokozeki et al. [152] in 2007 by studying solubility data of ammonia in different ionic liquids, a general principle has been established that low-temperature and high-pressure conditions are effective for absorption of ammonia, and therefore, would demand minimal requirements for auxiliary equipment for using in the synthesis loop of the Haber-Bosch process. A list of review works summarizing the prospects of ammonia removal by various ionic liquids has been comprehensively discussed and reported by Zhang et al. [153] for further reference.

3.4. Process intensification via integrated ammonia synthesis and separation

An aspect of process intensification is process engineering innovations by integration of multiple operations in a single unit to enhance the performance of a particular process objective [154,155]. In the Haber-Bosch process, the process intensification concept has brought about the idea of designing a multifunctional reactor that integrates the reaction and absorption units to not only improve the reaction performance but also enhance the overall energy efficiency of the process, on top of achieving reduced reactor size and minimal capital expenditure. In this process, high-pressure reaction and absorption, followed by a low-pressure desorption step. This idea introduces an unsteady-state process, where optimization is challenging due to conflicting temperature requirements of two distinctive stages of the ammonia production and separation within the single vessel. Using the existing commercial Fe-based catalysts, the reaction step requires high temperatures, while absorption benefits from lower temperatures. Additionally, reducing the temperature during desorption can lead to decreased efficiency [139].

In the work by Ojha et al. [141], the reaction and absorption processes were integrated to take place in a single vessel at 400°C . Conventional Fe-based catalysts were combined with NiCl_2 and MgCl_2 sorbents in a multifunctional reactor, where the vessel was closed but with fresh feed containing stoichiometric amounts of nitrogen and hydrogen gases being added to keep the pressure constant. To materialize this process intensification strategy, they introduced a semi-batch operation mode. The feed gases reacted to produce ammonia on the catalyst, followed by product gas desorption from the surface of the catalyst and then diffusion to the absorbent. Upon saturation of the absorbent, the ammonia released by pressure reduction. This was done by opening the vessel, allowing the gases to escape. The gases included

unabsorbed ammonia and the unreacted nitrogen and hydrogen. This concept, however, eliminated the recycling stream, demonstrating poor recovery and utilization of remaining valuable hydrogen and nitrogen gases. The only advantage of this proposed concept is the capital cost reduction by removing one unit operation.

Relatively recently, Smith and Torrente-Murciano [156] reported an investigation on integrated ammonia synthesis and absorption in a single unit by combining a newly formulated Ru-based catalyst and a newly developed MnCl_2 absorbent supported on silica gel in an attempt to achieve conversions above equilibrium in a single pass. They first developed the kinetic models of the catalyst and absorbent individually to predict the experimental long-term behavior, which were then used to facilitate the design of optimal operating conditions of the integrated system that could maximize the overall hydrogen conversion in the re-defined Haber-Bosch process. The integrated experimental setup and the arrangement of catalyst-absorbent layers used are depicted in Fig. 9. The temperature range was selected in such a way that the same temperatures for an effective catalytic activity overlapped with ammonia absorption that allowed achieving conversions beyond the reaction equilibrium. Upon optimizing the operating parameters that included the temperature, gas flow rate, and $\text{N}_2:\text{H}_2$ ratio, the mathematical modeling of both the kinetics of the catalyst and absorbent revealed that the size of the system, determined by the number of catalyst/absorbent layers, was dependent on the kinetics and equilibrium of the catalyst, while the ammonia production rate was dependent on the absorbent effectiveness. Operating the system using an $\text{N}_2:\text{H}_2$ ratio greater than stoichiometric was proposed to enhance the catalyst kinetics.

In addition to the integration of the ammonia synthesis unit with the absorption unit, the idea of process intensification can also be applied to design catalytic membrane reactors, where the packed-bed ammonia synthesis reactor is combined with a gas separation membrane in a single vessel. There have been extensive advancements in the development of membrane materials [22]. In such a scenario, the selective removal of a product takes place, effectively shifting the thermodynamic equilibrium and promoting greater product selectivity [157]. A review article by Moradihamedani [158] that presents recent developments in membrane technology for the elimination of ammonia from wastewater can be resourceful as preliminary literature for the adoption of this technology in the Haber-Bosch process.

As far as the literature is concerned, there have been limited numbers of experimental investigations conducted in studying the potential of this intensified process system to demonstrate high ammonia permeance and selectivity over nitrogen and hydrogen. In 2006, Camus et al. [159] performed experimental screening of MFI zeolite membranes in tubular and multi-channel fiber configurations, and tubular silica membranes at steady state to find a suitable membrane configuration to evaluate the effectiveness of ammonia recovery from syngas loop. It was found that reasonably high ammonia permeances were demonstrated for the tubular zeolite and silica membranes. In a relatively recent study by Wei et al. [160] in 2021, LTA-type zeolitic imidazolate framework membranes were investigated for the separation of ammonia, from nitrogen and hydrogen, reporting ammonia permeances as high as 1727 GPU (gas permeance unit), with selectivities of NH_3/N_2 and NH_3/H_2 of 35 and 12, respectively. The preferential adsorption of ammonia was enabled by the polar-polar interaction between ammonia and the linker polar channels for both gas pairs, and hence, thermodynamically favoring the separation of ammonia from nitrogen and hydrogen. A computational study reported by Zhang et al. [6] revealed an ammonia permeance of 100 GPU and selectivity of 10 over nitrogen and hydrogen, relaxing the thermodynamic constraint of the ammonia synthesis. The authors recognized the fact that not only this new design configuration still presents a considerable challenge, but the ammonia-permeable membranes are also still under development.

Taking a different approach, ion-exchange membranes are employed to produce ammonia in a membrane reactor. The majority of research has concentrated on the decomposition of ammonia using Pd-based

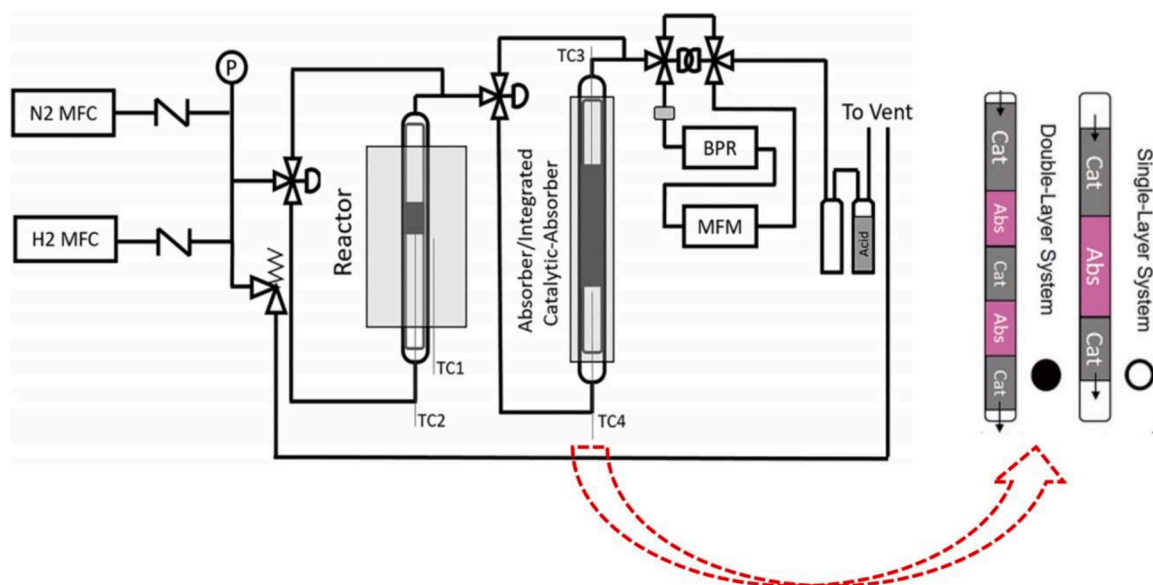


Fig. 9. Schematic of an integrated ammonia synthesis and absorption process, and the catalyst+absorbent configurations [156]. Reproduced from Wiley-VCH GmbH.

membranes [161], with relatively few studies dedicated to ammonia production. In ammonia production using a membrane reactor, two approaches involving either nitrogen or hydrogen-permeable membrane reactors exist. The utilization of a nitrogen membrane reactor for ammonia synthesis has been reported at 450°C and under ambient pressure [162]. The reaction involves N^{3-} ions diffusing through the nitride Nb_4N_5 membrane and reacting with H_2 . The difference in chemical potentials resulting from varying N_2 partial pressures serves as the driving force for enhancing the activation of N_2 , ion transport, and ammonia synthesis. For the hydrogen-permeable membrane, protons permeate electrochemically [163].

The concept of process intensification is not limited to integrating two or more processes in a single unit. Any modification made to reduce the size of a process unit, which subsequently increases the process throughput or improves the efficiency of the energy consumed hence decreasing the capital and operating expenditures, is considered a process intensification initiative [154]. Kale et al. [23] brought up the idea of redesigning the absorbent beds to enhance their thermal diffusivity by coating the absorbent onto the pore walls of a monolithic inert material. Similarly, the idea can also be extended to the redesigning of the ammonia catalyst beds using monolithic support structures. A monolithic structure is a single structure, through which the gas flow distribution is designed to contribute to enhanced mass and heat transfer effectiveness and reduced pressure drop operation [154,164–166]. The topology optimization method was employed to design a monolithic catalyst structure [167,168], aimed at enhancing the reaction rate of the system while limiting the pressure drop induced by the structure. Nonetheless, this new idea has not yet been demonstrated in industrial applications.

3.5. Purge gas recovery

To maintain a stable operation and prevent the gradual dilution of reactants due to the build-up of non-reacting gases, it is crucial to extract a purge stream. The composition of a standard purging gas generally comprises 60–85 % feed gas, 10–30 % inerts (mostly argon and methane), and a small percentage of ammonia [101]. Typically, ammonia in the purge is recovered through either condensation or by scrubbing it with water.

In recent studies, ionic liquids have emerged as an effective medium for ammonia recovery. These liquids are characterized by their unique

thermal, physical, and chemical properties [169], such as remarkable stability and minimal vapor pressure. This makes them excellent candidates for producing high-purity ammonia, as their low volatility significantly minimizes solvent evaporation during the gas separation process. Additionally, the use of ionic liquids lowers energy consumption, owing to their low heat capacity, especially when compared to water-based processes [153]. Chen et al. [170] proposed the utilization of [Bmim][BF₄] ionic liquid as an absorbent in a simulated ammonia absorption and flash recovery process using Aspen Plus software. This method was demonstrated to be more efficient in terms of solvent loss and energy use compared to traditional water washing techniques. A pilot plant in Sichuan, China, utilizing ionic liquid purification technology for ammonia recovery, has proved effectiveness with over 99.5 % ammonia removal. The high selectivity of the ionic liquid absorbent towards ammonia remained unaffected by the presence of hydrogen and nitrogen in the mixed gas [153]. The residual gas, which contains methane and hydrogen, can be utilized as a fuel in the reformer upstream. The hydrogen and nitrogen from the purge gas can also be recycled through different processes to serve as valuable raw materials for ammonia production.

In membrane separation, generally, a system comprised of hollow fibers is employed to separate hydrogen from the purge gas. The first industrial application of this technology [171] was launched by Monsanto in 1980. One notable drawback of membrane processes is that the pressure drop occurs on the hydrogen-rich gas side [101]. Optimal hydrogen recovery with minimal recompression costs is attained by implementing a two-step process with varying pressure ratios [172]. In the initial step, the feed contains approximately 62 % hydrogen, allowing for a 90 % hydrogen yield even at a relatively low-pressure ratio of 1.9. However, in the subsequent membrane, a higher pressure ratio is necessary to effectively separate hydrogen from a feed with a 30 % hydrogen concentration. Membranes are generally employed in cases where very high purity levels are not essential and their suitability hinges on factors such as feed pressure and the predominant impurities in the process [173].

PSA proves to be an appealing method for purge gas hydrogen recovery within ammonia plants due to its ability to achieve a very high hydrogen recovery rate, up to 99.999 % [173], while eliminating inert components. A key advantage is that pressure loss within the unit occurs on the off-gas side, while the hydrogen is readily available at nearly the same pressure as the feed gas. An economic assessment revealed that the

cost of waste hydrogen from ammonia plants to achieve a 99.97 % purity level is cost-effective and ranges between 1.17–1.39 € per kg of hydrogen [174].

Within the cryogenic separation process, the purge gas undergoes a scrubbing process with water to eliminate ammonia, followed by drying using molecular sieves to prevent any potential freezing of ammonia or water during the cryogenic process. In the subsequent stage, hydrogen is isolated through the condensation of inert components [101]. This approach demands higher capital investment and energy consumption compared to alternative methods [175].

Purge gas is also suggested as an effective argon source [176], and for its recovery, a complementary two-column cryogenic process has been proposed. This method targets argon extraction from hydrogen-depleted purge gas, and optimal operational parameters have been recommended to enhance efficiency [177].

4. Concluding remarks

In this literature review, we have examined the current state of process intensification in ammonia production, focusing primarily on the Haber-Bosch process. Initially, we explored alternative ammonia synthesis methods, establishing that the Haber-Bosch process is likely to remain the predominant method in the near future.

We observed that various companies employ different loop configurations in their process designs. Advancements in reactor design are predominantly undertaken by leading companies in the field, with limited publicly available research. Exergy analysis indicates that there are opportunities for improvement in the efficiency of the waste-heat-steam-generator and the main compressor. Various methods for recovering valuable gases from the purge gas are employed in the industry, tailored to specific purity and content requirements while ionic liquids have emerged as an effective medium for ammonia recovery.

Efforts to identify metal-based catalysts for dinitrogen reduction to facilitate ammonia synthesis at lower temperatures have spanned over a century. Yet, the catalysts used in today's industrial processes closely mirror those from the mid-1900s, primarily Fe and Ru-based. Despite several studies covered in recent reviews, which discuss preparation methods, promoter effects, and performance, substantial improvements at an industrial scale have been minimal. Ongoing research aimed at enhancing catalyst activity at lower temperatures may lead to paradigm shifts in ammonia synthesis. This could involve a transformative shift towards low-temperature and low-pressure operations, offering substantial process intensification with far-reaching implications. Such advancements could lead to smaller reactor and heat exchangers, reduced energy losses, and lower capital and operational costs. Particularly transformative would be the potential elimination of the main feed compressor, fundamentally altering ammonia production economics by enabling direct feed from an electrolyzer and significantly reducing the size of the recycle compressor. Additionally, for plants utilizing hydrogen sourced from an electrolyzer, water represents the only impurity requiring separation. This eliminates the necessity for a purge gas, thereby simplifying the process further.

Another promising area of research in process intensification lies within the ammonia separation process, presenting a viable alternative to the energy-intensive cryogenic methods. Recent studies have demonstrated various adsorptive methods for ammonia separation, but these methods generally show lower effectiveness compared to absorptive techniques. Absorbents are more promising for future exploration in ammonia separation technologies due to their higher capacity, better ammonia selectivity, and greater stability at high temperatures, as they selectively incorporate molecules into their crystal lattice. The employment of supported metal halides provides a significant advantage in terms of capacity and selectivity for ammonia at relatively high temperatures and low pressures. The primary challenge lies in the diffusion of ammonia into and out of the absorbents, marking the efficiency of ammonia absorption depending on the development of

materials capable of rapid ammonia uptake and release, while also maintaining stability under operational conditions. In the context of the Haber-Bosch process, which relies on absorption for ammonia separation, a crucial factor for enhancing energy efficiency is the convergence of temperatures for ammonia synthesis and absorption. This underscores the necessity for more active catalysts on the synthesis side and absorbents capable of higher uptake at higher temperatures on the separation side. Further efficiency gains could be achieved by harnessing the heat generated during absorption and ammonia synthesis and utilizing it in the desorption phase.

The concept of process intensification in the Haber-Bosch process has led to the idea of multifunctional reactors that merge the reaction and absorption phases to enhance reaction performance and energy efficiency while reducing reactor size and capital costs. However, this innovation introduces challenges in optimization due to the varying temperature needs of ammonia production and separation within the same unit. Furthermore, the application of process intensification has extended to the design of catalytic membrane reactors, combining ammonia synthesis and gas separation in one vessel. Advances in membrane material technology have facilitated this, allowing for the selective removal of products, thereby shifting the thermodynamic equilibrium towards greater product selectivity. Despite these advancements, experimental investigations into the effectiveness of these intensified systems in achieving high ammonia permeance and selectivity are still limited.

Other innovative methods for process intensification are being explored, including the use of monolithic structures as supports for catalysts or absorbents. To further enhance the capabilities of these structures, topology optimization has been proposed in the literature.

This review emphasizes the current research and development efforts across all aspects of the Haber-Bosch process. These advancements have the potential to create a process that is less energy-intensive and requires lower capital costs. When combined with a source of green hydrogen, these improvements could facilitate the decentralization of ammonia production. This would make ammonia a more accessible and sustainable option for energy storage and as a carrier in the future.

CRediT authorship contribution statement

Navid Erfani: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Luqmanulhakim Baharudin:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Matthew Watson:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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